MSG No. 93.126 P_I4_222 [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,7,8,9,12,13,14]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,3,7,8,9,12,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,12,13,14]`
4	$[0, 0, 0]$	`[10,11,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,4,5]`
3	$\left[0, 0, \frac{3}{4}\right]$	`[9,14,15,16]`
4	$\left[0, 0, \frac{1}{4}\right]$	`[10,11,12,13]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,6,10,11]`
2	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[2,3,9,14]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[4,5,15,16]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[7,8,12,13]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[4,5]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,8]
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[9,14]
6	$[0,\ 0,\ z]$	[10,11]
7	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[12,13]
8	$[0,\ 0,\ -z]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	[2,13]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[3,12]
4	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	[4,11]
5	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	[5,10]
6	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	[6,16]
7	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	[8,14]

Table 8: Wyckoff site: 8h, site symmetry: .2.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4]
2	$\left[0,\ x,\ \frac{1}{2}\right]$	[2,7]
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	[3,8]
4	$[-x,\ 0,\ 0]$	[5,6]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,12]
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[10,15]
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[11,16]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14]

Table 9: Wyckoff site: 8i, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,4]
2	$[0,\ x,\ 0]$	[2,7]
3	$[0,\ -x,\ 0]$	[3,8]
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[5,6]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[9,12]
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[10,15]
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[11,16]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[13,14]

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[2,5]`
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[3,4]`
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	`[6,8]`
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[9,15]`
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[11,12]`
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[14,16]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	`[3]`
4	$[x,\, -y,\, -z]$	`[4]`
5	$[-x,\, y,\, -z]$	`[5]`
6	$[-x,\, -y,\, z]$	`[6]`
7	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[7]`
8	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	`[11]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[14]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	`[16]`